

AI Value Sprint Canvas

Turn a business challenge into a small, measurable and responsible AI pilot.

TEAM / COOPERATIVE

Deep Learning Team Project (Group 4)

CASE / AREA

Viticulture / Vineyard Inspection

DATE

19/09/2026

1Business challenge

Vineyard managers have limited staff and time to decide which of thousands of vines need an in-person inspection. Undetected Black Rot, Esca or Leaf Blight can reduce yield and quality across Indian grape regions; unnecessary inspections wasting expert time.

2Evidence from reality

- GVLiD: 3,477 expert-verified RGB leaf photos, 4 classes, Tasgaon (Maharashtra), 2023.
- Only 2,407 are unique: 1,070 records were exact duplicates or conflicting labels.
- Black Rot fell from 808 to 97 images.
- Strong variation in light, background and scale.

3People involved

- Vineyard managers: set inspection priorities and routes.
- Field workers: photograph leaves (or in-situ cameras do it).
- Agronomists: review flagged leaves, inspect in-situ, decide.
- Data science team: build, validate and monitor the CNN.

4Opportunity question

How might we use vineyard photographs to prioritise which vines receive human inspection and follow-up, while keeping the expert in control?

5AI-enabled move

A CNN (3 convolutional blocks, trained from scratch, 128×128 input) returns a probability for each of the four classes per leaf photo. It is a triage signal, not a diagnosis: exact disease naming is not yet reliable. Training-only augmentation (flip, rotation, zoom) and class weights mitigate the imbalance. Next: transfer learning.

6Data and sources

- GVLiD, GrapeVine Leaf ID of Diseases (Shikalgar et al., 2026; DOI 10.17632/wkymf8bhcg.5).
- After SHA-256 deduplication: Healthy 1,093 (45%), Esca 886 (37%), Leaf Blight 331 (14%), Black Rot 97 (4%).
- Stratified 70/15/15 split; no exact duplicate appears in two sets.

7Human workflow

1. Employee photographs leaves on routine rounds (or in-situ cameras do).
2. CNN returns disease probabilities as a triage signal.
3. Agronomist reviews flagged leaves first and inspects in-situ.
4. Agronomist decides: monitor, sample or treat.
5. Outcome is logged for retraining.

8Trust and controls

Agronomist reviews every flag and inspects in-situ; no automated diagnosis or treatment; nothing auto-cleared.

Risks:
Leaf Blight recall 44% (27 of 50 called Esca), Black Rot 33% (15 test images); 6 Esca leaves missed as Healthy; folder labels are provisional; errors up to 92% confident; no unseen vineyard test.

9Test Horizon

Now: baseline CNN, 84.8% accuracy (majority class 45%), macro-F1 0.69; disease-vs-healthy recall 97%, 54% of test leaves flagged.

Next: transfer learning, confidence thresholds, new-vineyard pilot.

Go/No-Go: limited pilot; Go if macro-F1 > 0.69 and disease recall ≥ 97%.

★Value to create

Focus limited inspection time on the vines most likely to need expert attention. The CNN flags diseased leaves with 97% recall, so agronomists start with the highest risk, aiming to cut scouting time while keeping every decision.

Decision → Evidence → AI move → Responsible action